**TRIBHUVAN UNIVERSITY**

**INSTITUTE OF ENGINEERING**

**THAPATHALI CAMPUS**

**A MAJOR PROJECT FINAL REPORT**

**ON**

**"EVALUATING THE EFFECTIVENES OF VISVAP-PROGRAMMED SEMI-ACTUATED SIGNAL CONTROL OVER EXISTING FIXED-TIME CONTROL: A CASE STUDY OF BABARMAHAL INTERSECTION, KATHMANDU VALLEY "**


**Submitted By:**

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**Submitted To:**

Department of Civil Engineering

Thapathali Campus

Kathmandu, Nepal

**Under the Supervision of**

Dr. Rojee Pradhananga


February 2025

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**ABSTRACT**

150-300 words, include: Problem context, Approach, Key results, Impact. Avoid citations, unexpanded acronyms, detailed implementation.

***Keywords***—Combinatorial optimization, constraint satisfaction, genetic algorithm, hyper-heuristics, metaheuristic search, NP-hard problems, reinforcement learning, university course timetabling problem.

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# 1. INTRODUCTION

## 1.1. Background

Urban Traffic Congestion at signalized Intersection remains a Critical challenge in modern transportation system. Traffic congestion is the degradation of traffic flow condition on road and become slower speed which increase the travel time and increased Vehicle queues. It occurs when the demand of road space exceeds its capacity. Intersection Congestion is the main problem in urban area occurs when vehicle demand exceeds the capacity of a signalized or unsignalized crossing, leading to delays, queuing, and reduced traffic flow efficiency. Unlike general intersection congestion is often caused by conflicting vehicle movements, inefficient signal timing.

Congestion arises when there are more vehicles on the road than the infrastructure can handle. Significant delays in urban areas frequently originate at intersections, where conflicting traffic streams converge. There is extensive experience with chronic gridlock and congestion in the Kathmandu Valley. (Tiwari et al., 2023).

One of the popular concepts in traffic signalization is signalization is Fixed time traffic signal. A Fixed- time traffic signal is a traditional signal control system that operates on predetermined, unchanging timing plans. These signal cycle through preset green, yellow, and red phase at regular interval, regardless of real traffic demand.

Fixed time Traffic signal is wastes of times on empty approaches, which increases the delays in peak hours. When traffic demand fluctuated, the efficacy of Optimum Fixed-Time Management declined, resulting in significant increases in average vehicle delay, fuel consumption, and exhaust emissions. To solve the wastes of times on empty approaches, we introduce the actuated traffic signal. An actuated traffic signal is a signal control system that adjusts its timing in real-time based on vehicle or pedestrian demand, detected through sensor (e.g. inductive loops, camera, and radar).

A semi-Actuated Traffic signal is a type of Traffic Control system that combines elements of both fixed-time and fully actuated signals. It is mostly designed for the intersection where the major road carries a significantly higher traffic volume most of the time, while intersecting road i.e. minor road have experience lower, variable traffic flow. Semi-actuated signals operate with vehicle detection only on the minor approaches. The major road usually receives a consistent green phase by default. Vehicle arriving on the minor road are detected by sensors (like inductive loops or cameras).

Microsimulation tools like PTV VISSIM with VISVAP are widely used to modal and optimize actuated signal control. VISSIM is best software to simulation capabilities enable precise evaluation of traffic dynamics, while VISVAP facilitates the programming of adaptive signal logic. VISSIM simulation program, the time intervals when the fluctuations in traffic demands occur and the quantities (proportions) of the fluctuations at these time intervals can be determined by the users. Thus, changes in intersection performance resulting from short- or long-term fluctuations caused by concerts, sports events, traffic accidents, or adverse weather can be analyzed using VISSIM. (Cakici et al., 2021) .

### 1.2. Research Gaps and Problem Statement

Traffic congestion at major intersections throughout the Kathmandu Valley has reached critical levels, significantly degrading urban mobility, the economy, and public well-being. The current situation at the Babarmahal intersection is at a critical phase. The current fixed-time traffic signal system operates inefficiently, failing to adapt to the highly asymmetric traffic flows observed at this junction. The main issue of this intersection is the mismatched allocation of green signal time relative to actual Vehicle demand. The two major legs carry a high traffic volume, i.e., Baneshowr – Babarmahal leg and Maitighar-Babarmahal, whereas the two minor legs, i.e., toward Department of Road and toward Arts Gallery, carry a low traffic Volume. The inefficient Traffic light signal, i.e., long green time interval to the minor legs during periods when these minor legs experienced little or no Vehicular flow. Valuable green time is wasted on this time, which leads to heavily congested major roads and increases delay time.

The absence of the demand-responsive signal control is the main problem of this intersection. While a semi-actuated traffic signal system, which dynamically adjusts minor leg green time based on real-time vehicle detection , i.e., changes the traffic light toward the minor leg and gives the green on the major leg while there are no vehicles on the minor leg.

### 1.3. Objectives

This study aims to:

1. Assess the performance of a four-legged signalized intersection under existing fixed-time control using PTV VISSIM.
2. Optimize signal timings via actuated control in VISVAP, focusing on parameters like detector placement and green time allocation.
3. Compare the results with fixed-time control to quantify improvements in delay and queue length.

### 1.3.1. Study Area

The study focuses on the critical Babarmahal Road intersection (27.692506N,85.323863E) located in the administrative and cultural heart of Kathmandu, Nepal. It is a four-legged signalized intersection. The intersection features four distinct legs:

a. Major East leg: Carry a high-Volume Traffic along the Baneshwor-Babarmahal axis.
b. Major West leg: Carry a high-volume traffic along the Maitighar- Babarmahal axis.
c. Minor North leg: Leading directly towards the Department of Road (DOR).
d. Minor south leg: leading towards the Arts Gallery.

Figure 1-1. Aerial view of Intersection from Google Earth Pro

Figure 1-2. Intersection during visit

The traffic volume along the Maitighar (central) approach and the Baneshwor approach is significantly higher, classifying them as major roads within the intersection. In contrast, the Arts Gallery approach and the Department of Roads approach experience relatively lower and more fluctuating traffic volumes throughout the day, making them minor roads. Given this variation in traffic demand, implementing a semi-actuated traffic signal control system at this intersection is expected to be beneficial. Such a system can help reduce queue lengths, minimize vehicle delays, improve the Level of Service (LOS), and offer other operational advantages.

**Geometric Specifications of Babarmahal Intersection**

Table 1-1. Geometrical Parameters

| **Parameters** | **Specifications** |
|---|---|
| Number of Legs | 4(Four-legged intersection) |
| Approaches | Maitighar, Baneshwor, Arts Gallery, Department of Road (DOR) |
| Number of lanes Per Approach | Maitighar Approach: 8 lanes<br>Baneshwor Approach: 8 lanes<br>ArtsGallery Approach: 2 lanes<br>DOR Approach: 2 lanes |
| Lane Width | **Maitighar Approach:**<br>Service Lane: 3.5m(Total 4 lanes)<br>Main Lane: 3m(Total 4 lanes)<br>**Baneshowr Approach:**<br>Service Lane: 3.5m(Total 4 lanes)<br>Main Lane: 3m(Total 4 lanes)<br>**ArtsGallery Approach:**<br>Main Lane: about 2.5m(Total 2 lanes)<br>**DOR Approach:**<br>Main Lane: 2.5m(Total 2 lanes) |
| Pedestrian Crosswalks | Present on two legs (Maitighar approach and Arts Gallery approach) |
| Crosswalk Length | About 3.1m |
| Existing Signal Poles | Present at all four corners (Fixed-Time Signal Control) |

NOTE: The above measurements are preliminary estimates obtained using Google Earth Pro and will be verified through field surveys.

## 1.4. Research Questions

## 1.5. Limitations

## 2. LITERATURE REVIEW

Intersections are key junctions in the road networks where multiple roads meet or cross. Their design and traffic control methods play a vital role in ensuring smooth traffic flow, safety, and overall efficiency. Traffic management at intersections has been a complex issue in transportation systems for a long time. Several real-time signal control strategies have been implemented, but have not been able to effectively solve this problem completely (Tomar et al., 2022).

Broadly, intersections are categorized into signalized and unsignalized types based on the control mechanism used to manage traffic movement.

### 2.1. Signalized Intersection

A signalized intersection uses traffic signals to control the movement of vehicles and pedestrians. These signals show green, yellow, and red lights to indicate when to proceed, slow down, or stop. The signals operate on a timed cycle, assigning specific green light durations to different directions to maintain smooth and orderly traffic flow. (Duraku & Boshnjaku, 2024). Traffic signals help organize the flow of vehicles and people at intersections, making sure everyone gets their turn to move safely. By giving clear directions on when to stop and go, they help prevent traffic jams and reduce the chances of accidents. This makes the roads safer and more efficient for both drivers and pedestrians. "The traffic signal timing plan is the key that assigns the required time based on the travel demand at the intersection and keeps the cycle length to a minimum green time". The signal controller works as the time setting device for the specific intersection (Fwha,2008). Traffic flow movements are controlled by traffic signals without any conflict points in the signalized intersections. The movements are indicated by green, yellow, and red signals for the vehicle and pedestrian. The road users in the traffic include passenger cars, heavy goods vehicles, large goods vehicles, cycles, pedestrians, public transportation, and motorbikes (Eom & Kim, 2020). Traffic s ignals are electrically operated control devices to avoid conflicts between vehicles and pedestrians (Ryder Richardson et al., 2005). The three main concepts that describe the traffic signal are the cycle, cycle length, and phase. The cycle is the complete sequence of all the phases. Cycle length is the duration of one complete cycle. The phase is the time interval during which a specific group of vehicle traffic movements is permitted or restricted at the signalized intersection. The signal control cycle has different phases, with each phase representing a combination of permitted movements for multiple lanes receiving the right of way simultaneously (HCM, 2000). The basic control is applied through FTSC, and ATSC strategies include the control of FATSC (Full Actuated Traffic Signal Control), SATSC (Semi Actuated Traffic Signal Control), and Adaptive Traffic

Signal Control (ADTSC). These signal systems adjust according to traffic needs using data from detectors (Duraku & Boshnjaku, 2024)

## 2.2. Pretimed signal control

Pretimed control, also known as fixed time signal control, is a signal control system in which the cycle length, phase plan, and phase times are preset to repeat continuously. In this system, a preset time is given to each movement every cycle, regardless of changes in traffic conditions. Pretimed signal control operates on a fixed schedule, where each movement (e.g., vehicular or pedestrian phases) is allocated a predetermined amount of time within a signal cycle regardless of real-time traffic demand. (FHA,2007). Based on historic data at the intersection, the signal timing values were calculated, and these values are programmed into the pre-timed controllers. The pre-timed signal control has many advantages. For example, since both the start and end of green are predictable, it can be used for efficient coordination with adjacent pre-timed signals. Also, as this system does not require detectors, its operations are immune to problems that arise due to the failure of detectors. But on the other hand, fixed signal time control becomes inefficient at isolated intersections where traffic arrivals are very low. Pre-timed control is ideally suited to closely spaced intersections, mostly found in downtown areas. Also, this system is a great option for those intersections where a maximum of three phases is required (Urbanik et al., 2015). In the absence of traffic fluctuations in traffic demands, optimum fixed timed management (OFTM) is widely used in isolated intersection approaches to manage traffic flow. Fluctuations in traffic demand decrease the efficiency of OFTM, which causes longer queue lengths, delay, fuel consumption, and exhaust emissions (Grether et al.). Changing the timing of traffic signals is a cheap and simple way to improve traffic flow and reduce congestion. (FHA, 2007). But traditional fixed-time traffic signals operate on preset schedules and often fail to respond effectively to changing traffic volumes, especially during peak hours (Costa et al.,2018).

## 2.3. Actuated signal control

Whenever there are short-term or long-term fluctuations in traffic demand at some intersection approaches, the effectiveness and efficiency of the Optimum Fixed Time Management (OFTM) system might decrease or can be adversely affected (Grether et al., 2011). Traffic management systems based on any created algorithm are referred to as vehicle-actuated signal management systems. A vehicle-actuated management system determines the green and red signal timings according to the current dynamics of vehicle arrivals and queuing information obtained by detectors placed on the lanes in the intersection approaches. (Akcelik, 1994). One of the applications of Intelligent Transportation Systems (ITS) is vehicle-actuated (traffic- actuated) management systems. The effectiveness of these systems depends heavily on the control logic. The selection of the most appropriate control parameters for the designed control logic determines the effectiveness and functions of the actuated systems. (Cakici et al., 2021). Vehicle-actuated controllers are also a type of adaptive system, but the term “adaptive” usually refers to an area traffic control system (ATSC) where signal timings at various junctions are changed together based on that real time traffic conditions. (Ravikumar, 2012). An ATCS is a complex system that involves extensive surveillance, such as pavement loop detectors, and infrastructure to connect with central and local controllers. However, these systems are highly efficient, as they include algorithms that adjust a signal’s split, offset, phase length, and phase sequences to control and minimize delays, ultimately reducing the number of stops (Stevanovic, 2010). The analysis results show that the developed vehicle-actuated management system can effectively adapt to fluctuations in traffic demand at signalized intersection approaches.

The system significantly reduces average vehicle delays, fuel consumption, and exhaust emissions. Implementation of a vehicle-actuated management system can greatly improve the performance of an intersection in the presence of fluctuating or varying traffic demand. (Cakici et al., 2021). The actuated signal controller has detectors on all the approach lanes. When a detector detects the vehicle, the information is transferred to the controller, and the controller will give the service needed for the road users. The current signal tage will reach its minimum green time if they do not have more vehicles, and then it will change to the next stage to give the right of way to the vehicles. Detectors are placed on the lanes to communicate with the controller, and they should be properly numbered. The detector numbering depends on the phase numbering or the type of signal controller used. Fully actuated signal systems adjust the traffic lights based on real-time data collected from all lanes at an intersection, making these systems the most flexible. Semi-actuated systems only collect data from some lanes, so they only gather partial demand information and make decisions based on that data. Fully actuated control works best at isolated intersections where there are fluctuations in traffic demand throughout the day. (Urbanik et al., 2015). The maximum green time should be set for the actuated signals to determine the green signal splits for the intersection. If the maximum green time is set too high, then the high-demand lanes would start to affect the operations on the other approach lanes. The minimum green time is the amount of green time required for the vehicle to clear the signalized intersections; it should be enough for the vehicle to pass the intersection, otherwise it will affect the efficiency of the intersection (Sinowatcher, n.d.).

In a semi-actuated control system, the design logic control dynamically adjusts signal timings by determining the green light duration for the secondary road based on the arriving traffic flow. When the detector detects no vehicles or the absence of traffic, the control logic eliminates the green interval and moves to the next phase, which greatly reduces cycle time and delay. This dynamic adjustment of the switching signal operates instantly using an 'if-then' conditional logic to optimize traffic signal performance (Duraku & Boshnjaku, 2024). Vehicle-actuated signal systems were developed to improve upon fixed-time systems by detecting vehicles approaching an intersection and adjusting green time accordingly.

Although more efficient than fixed-time systems, vehicle-actuated signals are still limited in their ability to coordinate across multiple intersections (Brown & Patel, 2019).

The cost for both the initial set-up and the maintenance is the major disadvantage of fully actuated control. The high cost is due to the amount of detection required. Also, there are chances for a higher percentage of vehicles to stop since the green time is not held for upstream traffic. Another disadvantage is that, if the traffic patterns are regular, the extra benefit obtained from the addition of local actuation is minimal or can be zero (Fhwa, 2008). Using a fuzzy logic-based vehicle actuated management system reduced delays by 10investigation for vehicle-actuated signal control (Wang et al., 2018). The green and red signal timings at each cycle were determined using global optimization and complementarity for vehicle-actuated management systems (Ribeiro and Simoes et al). At the end of the analysis, it was concluded that a fully-actuated management system could provide better and more encouraging results than the fixed- time and semi-actuated management systems.

### 2.4. Adaptive signal control

Adaptive Traffic Control Systems (ATCSs) change signal timings in real time based on how much traffic is on the road and how the system is working. These systems use smart rules to change things like how long lights stay green, when they start, and the order of signals to reduce delays and stops. They need a lot of monitoring, often using sensors under the road, and special equipment to connect with control centers(Stevanovic, 2010). Adaptive traffic control systems are developed to solve the limitations of pre-timed control by instantly adapting to field conditions, such as changing traffic signal timings based on the traffic present and fluctuations in demand or movement(Martin, 2008). Various adaptive traffic control systems like SCOOT, SCATS, and OPAC optimize signal timings based on real-time and predicted traffic demand to improve flow through coordinated green waves. However, the installation and maintenance costs of these systems are very high, and they also require complex infrastructure. As a result, they are not widely implemented and are built in only a few cities around the world (Tomar et al., 2022).

# 3. THEORETICAL BACKGROUND

## 3.1. Terminologies and Basic Definitions

Traffic flow patterns can be categorized into several types based on the movement and interaction of vehicles on the roadway. The main types include:

## 3.2. Traffic Flow

### 3.2.1. Types of Traffic Flow Pattern

### 3.2.2. Traffic Conflicts

## 3.3. Level of Service(LOS)

## 3.4. Capacity

## 3.5. Vehicle to Capacity Ratio

## 3.6. Headway

### 3.6.1. Time Headway

### 3.6.2. Distance Headway

## 3.7. Saturation Flow

## 3.8. Passenger Car Unit

## 3.9. Peak Hour Factor(PHF)

## 3.10. Unsignalized Intersection

## 3.11. Signalized Intersection

## 3.12. Control Delay

## 3.13. Signal Timing

## 3.14. Phase Design

### 3.14.1. Two Phase Design

### 3.14.2. Four Phase Design

## 3.15. Queing Theory

## 3.16. Traffic Simulation

# 4. METHODOLOGY

## 4.1. Study Area and Description.

## 4.2. Primary data collection.

The data used in this study were obtained from the District Police Office, Bhadrakali, Kathmandu, which provided CCTV video footage of the Babarmahal Intersection. The footage covered a continuous 24-hour period over three consecutive weekdays, from July 20th, 2025, to July 22nd, 2025, capturing the full spectrum of daily traffic conditions, including morning peak, off-peak, evening peak, and nighttime hours.

The use of CCTV-based video data ensured non-intrusive and accurate traffic observation without disrupting the natural flow of traffic at the intersection. The video footage was subsequently processed to extract classified traffic volume counts, turning movement counts, and other relevant traffic parameters required for the development and calibration of the microsimulation model in PTV VISSIM, as described in the subsequent sections.

The geometric data, including lane widths and GPS coordinates for each lane, were obtained through **field measurement and a GPS survey** conducted on June 5th, 2025. Signal timing parameters, including cycle length, phase sequences, and green times, were recorded from the existing traffic signal controller at the site.All collected data were subsequently used to develop and calibrate the microsimulation model in **PTV VISSIM**, as described in the subsequent sections.

The video footage was processed and analyzed to determine the peak and off-peak traffic conditions at the intersection. Traffic volume observation extracted from the footage revealed that **the maximum traffic flow was observed between 9:30 AM and 10:30 AM**, which was therefore selected as the **peak hour** for this study. Similarly, the period between **1:00 PM and 2:00 PM** was identified as representing comparatively lower traffic demand and was selected as the **off-peak hour**. Classified traffic volume counts and turning movement counts were extracted from the CCTV footage for both these one-hour periods and used as input data for the microsimulation model in PTV VISSIM.For the purpose of traffic volume count, vehicles were classified into eight categories, namely Heavy Truck, Mini Truck, Big Bus, Micro Bus, Utility Van, Pickup, Safa Tempo, and Bike, in accordance with the vehicle classification system adopted by the **Department of Roads, Nepal Road Standard 2070**.

## 4.3. Existing Conditions analysis using the HCM manual.

The performance of the Babarmahal Intersection under the existing fixed-time signal control was analytically assessed following the signalized intersection analysis procedure outlined in the Highway Capacity Manual (HCM). This manual method was applied to the peak-hour traffic data collected on the first day of the survey (9:30–10:30 AM) to determine approach-level and intersection-wide values of control delay and Level of Service (LOS). The HCM analysis serves as an independent analytical baseline for Scenario 1, allowing a validation comparison with the microsimulation results obtained from PTV VISSIM.

The existing signal operates on a cycle length of 157 seconds, comprising four phases. The green time allocations for each phase are summarized in Table 4-1

Table 4-1. Existing Fixed-Time Signal Phase Plan — Babarmahal Intersection

| Phase | Movement Description | Green Time (s) |
|---|---|---|
| Phase A | Right turns: Baneshwor → DOR and Maitighar → Art Gallery | 25 |
| Phase B | Through movement: Maitighar ↔ Baneshwor (main corridor) | 80 |
| Phase C | Art Gallery approach movements | 25 |
| Phase D | DOR approach movements | 27 |

#### 4.3.1. Step 1: Peak Hour Factor (PHF)

The Peak Hour Factor (PHF) quantifies how uniformly traffic is distributed within the peak hour. It is defined as the ratio of the total hourly volume to four times the peak 15-minute volume, and is given by:

$$\text{PHF} = \frac{V_{hour}}{4 \times V_{15,peak}}$$

where:

- $V_{hour}$ = the total PCU count during the peak hour
- $V_{15,peak}$ = the highest 15-minute PCU count within that hour

. PHF values closer to 1.0 indicate a more uniform distribution of traffic, while lower values indicate a sharply peaked flow pattern.

For the Maitighar approach, the 15-minute PCU totals across all turning movements (towards Baneshwor service lane, Baneshwor main lane, DOR, and Art Gallery) were summed for each interval as follows: 9:30–9:45 = 403.00; 9:45–10:00 = 386.25; 10:00–10:15 = 451.50; and 10:15–10:30 = 491.00. The total hourly volume is 1731.75

PCU/hr, with the peak 15-minute interval occurring at 10:15–10:30. Substituting into the formula:

$$\begin{aligned} PHF &= 1731.75/(4 \times 491.00) \\ &= 1731.75/1964.00 \\ &= 0.882 \end{aligned}$$

The PHF values for all four approaches were similarly computed and are presented in Table

| **Approach** | **9:30-9:45** | **9:45-10:00** | **10:00-10:15** | **10:15-10:30** | **Total (PCU/hr)** | **Peak $V_{15}$** | **PHF** |
|---|---|---|---|---|---|---|---|
| Maitighar | 403.00 | 386.25 | 451.50 | 491.00 | 1731.75 | 491.00 | 0.882 |
| Baneshwor | 621.50 | 533.75 | 490.25 | 538.75 | 2184.25 | 621.50 | 0.879 |
| Art Gallery | 92.25 | 85.00 | 85.75 | 89.50 | 352.50 | 92.25 | 0.955 |
| DOR | 5.50 | 12.25 | 15.00 | 20.50 | 53.25 | 20.50 | 0.649 |

### 4.3.2. Step 2: Peak 15-Minute Flow Rate

The hourly field volume V (PCU/hr) is converted to a peak 15-minute equivalent flow rate v using the PHF. This conversion accounts for the fact that traffic demand within the peak hour is not uniform, and the highest intensity period governs capacity and delay. The flow rate is given by:

$$v = V/PHF$$

For the Maitighar approach:

$$\begin{aligned} v &= 1731.75/0.882 \\ &= 1963.44 PCU/hr \end{aligned}$$

Dividing by a PHF less than 1.0 yields a flow rate higher than the raw hourly count, representing the worst-case demand intensity that the intersection must accommodate. The flow rates, along with the corresponding signal phase and green time, for all approaches are presented in Table X+2.

### 4.3.3. Step 3: Saturation Flow Rate

The saturation flow rate (s) represents the maximum number of PCUs that can be discharged from an approach per hour, assuming a continuous green signal. Under ideal

| Approach | V (PCU/hr) | PHF | v (PCU/hr) | Phase | Green (s) | Cycle (s) | g/C |
|---|---|---|---|---|---|---|---|
| Maitighar | 403.00 | 386.25 | 451.50 | 491.00 | 1731.75 | 491.00 | 0.882 |
| Baneshwor | 621.50 | 533.75 | 490.25 | 538.75 | 2184.25 | 621.50 | 0.879 |
| Art Gallery | 92.25 | 85.00 | 85.75 | 89.50 | 352.50 | 92.25 | 0.955 |
| DOR | 5.50 | 12.25 | 15.00 | 20.50 | 53.25 | 20.50 | 0.649 |

conditions, a single lane has a base saturation flow rate ($s_o$) of 1900 PCU/hr/ln. This base value is adjusted for prevailing field conditions using the following expression:

$$s = s_o \times f_w \times f_{HV} \times f_g \times f_a \times f_{LU} \times N$$

where:

- $s_o$ = base saturation flow rate
- $f_w$ = lane width adjustment factor
- $f_{HV}$ = heavy vehicle adjustment factor
- $f_g$ = grade adjustment factor
- $f_a$ = area type adjustment factor
- $f_{LU}$ = lane utilisation factor
- N = number of lanes

#### 4.3.4. Heavy Vehicle Factor ($f_{HV}$)

Among the adjustment factors, the heavy vehicle factor is the most significant for the Maitighar and Baneshwor approaches due to the considerable number of large buses observed, particularly in the service lanes. The factor is computed as:

$$f_{HV} = 1/[1 + P_{HV} \times (E_T - 1)]$$

where:

- $P_{HV}$ = is the proportion of heavy vehicles in the traffic stream
- $E_T$ = passenger car equivalent for heavy vehicles, taken as 2.0

For the Maitighar service lane, 131 out of 689 counted vehicles were big buses ($P_{HV}$ = 0.19), yielding a weighted average $f_{HV}$ = 0.9241 across both lanes combined. The adjustment factor values applied to each approach are summarised in Table X+3.

#### 4.3.5. Step 4: Capacity and Degree of Saturation

The effective capacity (c) of each approach lane group is the product of the saturation flow rate and the green ratio (g/C), representing the number of PCUs that can be served per hour given the available green time:

$$c = s \times (g/C)$$

The degree of saturation, or volume-to-capacity ratio (X), is then computed as the ratio of the peak flow rate to the capacity:

$$X = v/c$$

An X value below 1.0 indicates that the approach is operating within capacity. An X value exceeding 1.0 signals oversaturation, meaning vehicles cannot all be discharged within a single cycle, and residual queues accumulate progressively. For the Maitighar approach:

$$c = 3002.31 \times (80/157)$$
$$= 1529.84 \text{ PCU/hr}$$

$$X = 1963.44/1529.84$$
$$= 1.2834$$

The capacity and degree of saturation for all approaches are presented in Table X+4.

#### 4.3.6. Step 5: Control Delay

Control delay per vehicle is the primary performance measure for signalized intersections under the HCM methodology. It comprises three components:

$$d = d_1 + d_2 + d_3$$

#### 4.3.7. Uniform Delay ($d_1$)

Uniform delay assumes vehicles arrive at a perfectly steady rate throughout the cycle. It is given by:

$$d_1 = [0.5 \times C \times (1 - g/C)^2]/[1 - \min(1, X) \times (g/C)]$$

The term min(1, X) prevents the denominator from becoming negative or zero under oversaturated conditions. For the Maitighar approach:

$$d_1 = [0.5 \times 157 \times (1 - 0.5096)^2]/[1 - \min(1, 1.2834) \times 0.5096]$$
$$= 18.88/0.4904$$
$$= 38.50 \text{ sec/veh}$$

### 4.3.8. Incremental Delay ($d_2$)

Incremental delay accounts for the additional waiting caused by random vehicle arrivals and, more critically, by residual overflow queues when demand exceeds capacity. It is expressed as:

$$d_2 = 900T \times [(X-1) + \sqrt{((X-1)^2 + (8 \times kIX/cT))}]$$

where:

- T = analysis period duration (0.25 hr for a 15-minute period)
- k = incremental delay factor (0.5 for fixed-time signals)
- I = upstream filtering factor (1.0 for isolated intersections)

For the Maitighar approach:

$$\begin{aligned} d_2 &= 900 \times 0.25 \times [(1.2834-1) + \sqrt{((1.2834-1)^2 + \frac{(8 \times 0.5 \times 1.0 \times 1.2834)}{(1529.84 \times 0.25))}}] \\ &= 225 \times [0.2834 + \sqrt{(0.0803 + 0.01342)}] \\ &= 225 \times [0.2834 + 0.3062] \\ &= 225 \times 0.5896 \\ &= 132.66\ sec/veh \end{aligned}$$

### 4.3.9. Initial Queue Delay ($d_3$)

The initial queue delay $d_3$ accounts for any pre-existing residual queue at the start of the analysis period. Since the analysis begins at the start of the peak hour with no assumed carry-over queue from a prior period, $d_3$= 0 for all approaches.

### 4.3.10. Total Control Delay — Maitighar Approach

$$\begin{aligned} d &= d_1 + d_2 + d_3 \\ &= 38.50 + 132.66 + 0 \\ &= 171.16\ sec/veh \end{aligned}$$

At 171.16 sec/veh, the Maitighar approach operates at Level of Service F, indicating a breakdown condition where demand substantially exceeds the signal's serving capacity, causing progressive queue accumulation across cycles. The complete delay results for all approaches are given in Table X+5.

### 4.3.11. Step 6: Intersection-Wide Delay and Level of Service

### 4.3.12. Key Observations

### 4.3.13. Comparison of HCM and VISSIM Results for Existing Scenario

### 4.3.14. Aggregation of VISSIM Movement-Level Results

PTV VISSIM reports queue delay for each individual turning movement within the intersection. To derive approach-level LOS values comparable to those from HCM, the movement delays were averaged for each approach and the resulting value was classified against HCM LOS thresholds. The procedure is illustrated using the Maitighar approach as a sample calculation.

Table 4-2. VISSIM Movement-Level Queue Delays — Maitighar Approach (Scenario 1)

| Movement | Queue Delay (sec/veh) |
|---|---|
| Maitighar Service → Anamnagar | 28.43 |
| Maitighar Service → Babarmahal Service | 45.05 |
| Maitighar Main → Art Gallery | 139.64 |
| Maitighar Main → Babarmahal Main | 51.40 |

The approach-level average queue delay is computed as:

$$\begin{aligned} AverageDelay &= (28.43 + 45.05 + 139.64 + 51.40)/4 \\ &= 264.52/4 \\ &= 66.13\ sec/veh \end{aligned}$$

Based on Table 4-2 HCM LOS thresholds, a delay of 66.13 sec/veh falls within the range of 55–80 sec/veh, corresponding to Level of Service E. The same procedure was applied to the remaining three approaches. The aggregated approach-level VISSIM results are presented in Table 4-3

### 4.3.15. HCM vs VISSIM — Comparison of Level of Service

Table 4-4 presents a side-by-side comparison of the Level of Service assigned by HCM and VISSIM for each approach and the overall intersection under the existing signal control scenario. As discussed above, LOS is used as the basis for comparison since it is derived from the same HCM thresholds in both methods, making it the most appropriate common metric.

Table 4-3. Aggregated VISSIM Approach-Level Queue Delay and LOS — Scenario 1

| Approach | Movements Averaged | Avg. Queue Delay (sec/veh) | LOS |
|---|---|---|---|
| Maitighar | 4 movements | 66.13 | E |
| Babarmahal | 4 movements | 66.28 | E |
| Art Gallery | 5 movements | 45.71 | D |
| Anamnagar | 3 movements | 57.93 | E |
| **Overall Intersection** | **(VISSIM node result)** | **59.60** | **E** |

Table 4-4. Comparison of HCM and VISSIM Level of Service — Existing Scenario (Scenario 1) Approach HCM Control Delay (sec/veh) HCM LOS VISSIM LOS

| Approach | HCM Control Delay (sec/veh) | HCM LOS | VISSIM LOS |
|---|---|---|---|
| Maitighar | 171.16 | F | E |
| Babarmahal | 301.91 | F | E |
| Art Gallery | 248.14 | F | D |
| Anamnagar | 58.89 | E | E |
| **Overall Intersection** | **241.39** | **F** | **E** |

#### 4.3.16. Discussion

The LOS-based comparison presented in Table X+2 reveals both areas of agreement and divergence between the HCM and VISSIM methods. The following observations are drawn from the comparison:

**Methodological Differences in Delay Parameters:** As noted earlier, HCM control delay and VISSIM queue delay are related but not equivalent parameters. HCM derives control delay analytically using deterministic formulas that assume idealized lane discipline, uniform vehicle behavior, and standardized saturation flow adjustment factors. VISSIM, on the other hand, measures queue delay empirically from simulated vehicle trajectories, incorporating field-calibrated driver behavior parameters that more realistically reflect the mixed traffic conditions and poor lane discipline characteristic of Kathmandu urban intersections. These methodological differences are the primary reason for the LOS divergence observed across most approaches.

**Anamnagar Approach — Agreement:** Both HCM and VISSIM assign LOS E to the Anamnagar approach, representing the strongest point of agreement between the two methods. This convergence is expected given that the Anamnagar approach operates well below capacity, where overflow queuing is minimal, and the deterministic HCM formula more closely approximates real conditions simulated by VISSIM.

**Qualitative Agreement on Overall Congestion:** Despite the LOS differences at individual approaches, both methods consistently identify Babarmahal Intersection as operating under severely congested conditions during the morning peak hour. HCM classifies three approaches at LOS F and one at LOS E, while VISSIM classifies the overall intersection at LOS E. The directional agreement — that the intersection is significantly over-stressed under the existing fixed-time signal plan — validates the general conclusion drawn from both methods and reinforces the need for signal timing optimization.

**Basis for Methodology Decision:** Given the fundamental differences between HCM control delay and VISSIM queue delay, and the inherent limitations of applying HCM's idealized assumptions to the complex mixed traffic environment of Kathmandu, VISSIM node results are adopted as the primary and consistent source of performance data across all three scenarios in this study. The HCM analysis serves as a supplementary analytical reference that corroborates the qualitative finding of poor intersection performance under the existing signal control.

### 4.4. Network Coding

Here we draw links, connectors and intersection geometry based on field data. Link represents a single or multiple road ways and connectors are used for joining the two links.

#### 4.4.1. Vehicle Input and Composition

We performed the manual traffic volume count along with vehicle composition for various approaches for different vehicles classes. By taking speed data of various classes of vehicles, we have done the speed distribution of different vehicle classes. The speed data was extracted from the CCTV footage by dividing the certain distance with time required for that vehicle class to travel that distance. Multiple observations were made and the manual speed data was taken and compared with the one taken from CCTV footage.

Firstly, traffic volume count was entered for various approaches. It was further divided into 4-time intervals with 15 minutes intervals each, but while entering in VISSIM we multiply it by 4 since it asks us for Number of vehicles per hour and not per time interval.

2D/3D models of various vehicle classes was taken from Sketch-UP and entered into the VISSIM with standard dimensions of the vehicle as Nepal Road Standard(2070).

Then we perform the vehicle composition where we divide the traffic input into different vehicle classes.

$$Vehicle\ Composition(Bike) = \frac{Traffic\ count\ of\ Bike}{Total\ traffic\ count} \text{(for specific purpose)}$$

#### 4.4.2. Routing Decisions (Turning Movements)

For each vehicle classes it was defined the proportion of vehicles turning left, going through, and turning right at each leg from Static Vehicle Routing Decision/ Static Vehicle Routes.

#### 4.4.3. Driving Behavior Parameters

Setting Wiedemann car-following model parameters and lane change behavior suited to mixed traffic conditions.

#### 4.4.4. Signal Controller Setup

Defining fixed-time signal phases, cycle length, green/amber/red times, and phase sequences.

#### 4.4.5. Conflict Areas and Priority Rules

Defining vehicle interaction zones within the intersection. The conflict areas were modelled using passive priority to replicate the non-lane-disciplined and gap-acceptance based driving behavior observed in study area. As the intersection operates under signal control, explicit priority assignment was not needed. The setting of passive conflict zones allowed smoother vehicle interactions and improved simulation stability that closely resembled the real driver behavior in congested urban intersections.

#### 4.4.6. Pedestrian Input

Adding pedestrian crossings and volume if considered in the study.

### 4.5. Calibration and Validation

The developed VISSIM model was calibrated and validated using the traffic data extracted from the CCTV footage obtained from the District Police Office, Bhadrakali. For the purpose of calibration, traffic volume data collected during two consecutive days (July 20th and July 21st) were utilized. The model parameters, including Wiedemann car-following model coefficients and desired speed distributions, were iteratively adjusted until the simulated traffic volumes and queue lengths sufficiently replicated the observed field conditions.

The calibrated model was subsequently validated using the traffic data from the third day (July 22nd), which was kept independent of the calibration process to ensure an unbiased assessment of model performance.

Calculation of GEH value: The GEH statistic is calculated using the formula:

$$GEH = \sqrt{\frac{2(V_s - V_o)^2}{V_s + V_o}} \tag{4-1}$$

Where:

- $V_s$ = Simulated volume (from VISSIM)
- $V_o$ = Observed volume (from field data)

Acceptance Criteria:

- $GEH < 5.0 \rightarrow$ Good fit
- $GEH$ 5.0 – 10.0 $\rightarrow$ Acceptable (investigate further)
- $GEH > 10.0 \rightarrow$ Poor fit (recalibrate)

### 4.6. Semi Actuated Control

Semi-actuated control is a practical traffic management strategy designed specifically for intersections where a heavily traveled major street crosses a minor street with much lighter traffic. In these situations, the main goal is to keep traffic flowing smoothly on the major street while still allowing minor street traffic to proceed when needed. To achieve this, the major street is given priority and holds the green light by default. The right-of-way is only transferred to the minor street when a vehicle needs to cross or merge onto the major street and cannot find a safe gap in traffic. Because the major street automatically receives the green light, there is no need to install detectors (sensors) on it. Instead, detection equipment is placed only on the minor street approaches to monitor incoming traffic. This approach is both cost-effective and efficient, as it focuses monitoring resources where they are actually needed.

Semi-actuated control works best under specific traffic conditions. It is ideal when there is a clear difference in traffic volume between the two intersecting roads. If an intersection has two roads of similar importance and traffic levels, a fully-actuated system would be better. However, when there is a significant difference in traffic volume, semi-actuated control prevents the unnecessary delays that fixed-time systems often create on busy major streets.

Semi-actuated control is efficient because it keeps the green light on the major street most of the time. This means fewer stops for drivers on the busy road. The system is also simple to use. Drivers on the minor street just stop at the line and wait. When the sensor detects them, the controller transfers the green light to let them cross. This way, minor street traffic is never completely stuck, and the major road still gets priority. The system also costs less to install and maintain because it only needs sensors on the minor street, not on the major street (Ni, 2020).

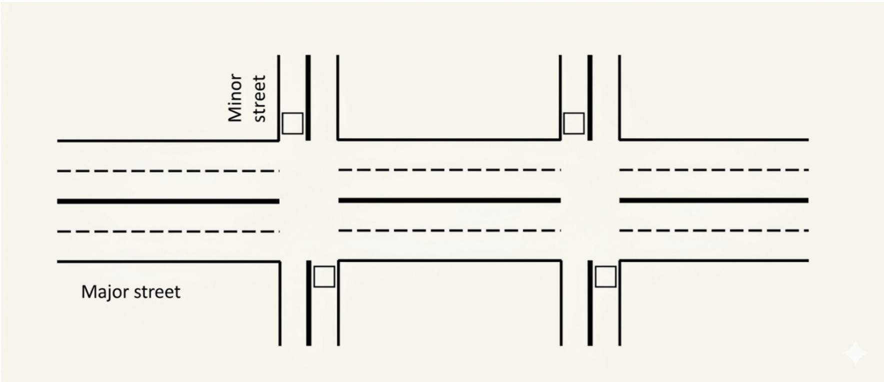


Figure 4-1. Application scenario of Semi-Actuated

### 4.7. Working Mechanism of Semi-Actuated Control

When the side street receives a green light, it is guaranteed a minimum amount of time to remain green. If no vehicles are waiting or present after this minimum time, the light simply stays green for that duration and then changes back to the main road. This natural and safe transition is called a "Gap Out." However, if vehicles are still present after the minimum time, the sensor detects them. For every new vehicle that passes over the sensor, the system adds a small amount of extra time, called "Passage Time," to allow it through. This process continues until the arrival of vehicles stops, and none arrive within that passage time window. At that point, the light safely "Gaps Out" to return control to the main road. If a large queue of vehicles accumulates on the side street, they could potentially extend the green light indefinitely, causing traffic congestion on the busy main road. To prevent this, the system enforces an absolute time limit called "Maximum Green." Once this limit is reached, the system abruptly terminates the side street signal and forces the light to change. This forced termination is called a "Max Out. "If a ‘Max Out’ occurs and leaves a vehicle trapped beyond the sensor, the system creates an ‘artificial call’ to remember that vehicle. This ensures the signal returns to the side street as soon as possible, allowing the stranded vehicle to proceed safely and effectively (Ni, 2020)

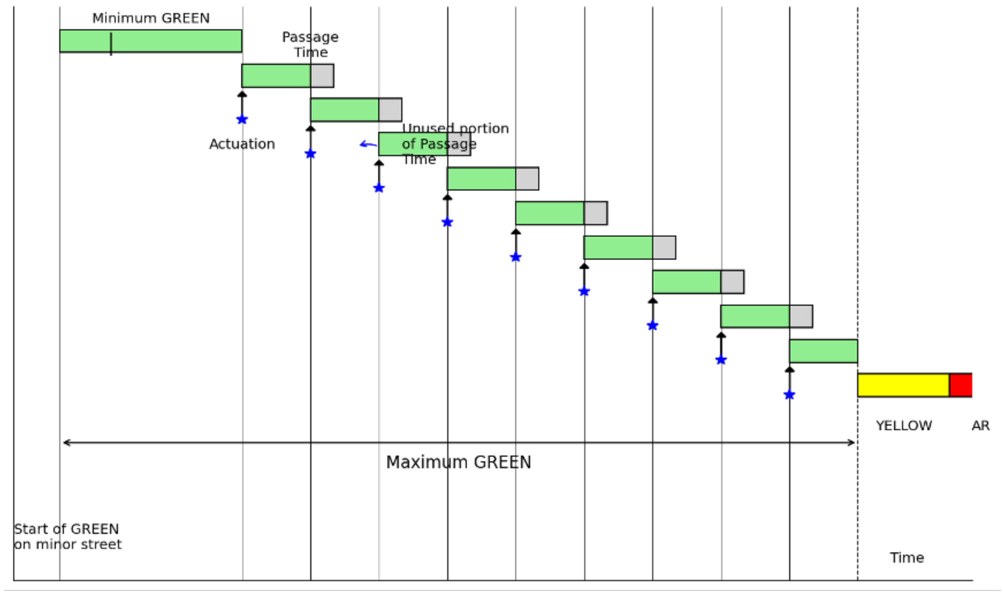


Figure 4-2. Working Principle of Semi-Actuated signal control

### 4.8. Basic Timing Parameters in Actuated Control

In Semi-Actuated Traffic Signal Control (SATSC), signal timings are dynamically adjusted based on vehicle detection. However, several fundamental timing parameters must first be calculated to ensure proper operation of the signal system. These include the minimum green time, extension time (unit extension), passage time, maximum green time, and cycle length. These parameters ensure that queues are cleared efficiently while maintaining flexibility for varying traffic demand (Duraku & Boshnjaku, 2024).

#### 4.8.1. Minimum Green Time ($G_{min}$)

The minimum green time represents the shortest duration for which a signal phase remains green once activated. This parameter ensures that vehicles queued at the stop line have enough time to begin moving and pass through the intersection. It also accounts for the start-up lost time caused by driver reaction and vehicle acceleration (Duraku & Boshnjaku, 2024).

The minimum green time is mathematically related to the detector placement and queue length. It is calculated using Equation (4-2):

$$G_{min} = T_L + [h * \text{Integer}(d/x)] \tag{4-2}$$

where:

- Gmin = Minimum green time (seconds)

- TL = Start-up time loss (typically 2.0 – 4.0 seconds)
- h = Saturation headway (seconds/vehicle)
- d = Distance between detector and stop line (m)
- x = Average spacing between stored vehicles (typically 6–7 m)

The integer function represents the number of vehicles stored between the detector and the stop line, which determines the minimum time required for queue clearance.

### 4.8.2. Extension Time or Unit Extension (U)

The extension unit (U) represents the additional time added to the green interval when a vehicle is detected during the green phase. This allows the signal to remain green as long as vehicles continue arriving within a specified time gap. According to the Traffic Detector Manual, the recommended extension time is (Duraku & Boshnjaku, 2024):

- 3.0 seconds for approach speeds ≤ 45 km/h
- 3.5 seconds for higher speeds

### 4.8.3. Passage Time (P)

Passage time is the travel time required for a vehicle to move from the detector location to the stop line. It is calculated using Equation (4-3):

$$P = \frac{d}{1.47 \times S_{15}} \tag{4-3}$$

where:

- P = Passage time (seconds)
- d = Distance between detector and stop line (ft)
- $S_{15}$ = 15th percentile approach speed of vehicles (mph)

To ensure proper operation, the condition U ≥ P must be satisfied, ensuring the green phase remains active long enough for the approaching vehicle to reach the stop line.

### 4.8.4. Maximum Green Time ($G_{max}$)

The maximum green time defines the upper limit for a green phase, preventing excessive delay for conflicting movements. The effective green time ($g_i$) for each phase is distributed proportionally based on traffic demand using Equation (4-4):

$$g_i = (C_i - L) \times \frac{V_{C_i}}{V_C} \tag{4-4}$$

Where:

- $g_i$ = Effective green time for phase i (s)
- $C_i$ = Cycle length (s)
- L = Total lost time in the cycle (s)
- $V_{C_i}$ = Traffic volume in the critical lane for phase i (pcu/h)
- VC = Sum of volumes in all critical lanes (pcu/h)

### 4.8.5. Critical Cycle Length ($C_c$)

The critical cycle length is the total time for all phases to complete one sequence when the side street reaches Gmax. It is calculated using Equation (4-5)

$$C_c = \sum (G_i + Y_i) \tag{4-5}$$

where:

- $C_c$ = Critical cycle length (seconds)
- $G_i$ = Maximum green time for phase i
- $Y_i$ = Yellow plus all-red interval for phase

### 4.8.6. Calculation of Parameters:

To demonstrate the application of signal timing parameters for the Babarmahal intersection, detailed VisVAP calculations are performed for the off-peak condition. The calculations follow the HCM methodology and incorporate sustainability principles for enhancing signal controller flexibility.

The total cycle length was set at 169 seconds, matching the current timing at the intersection and confirmed through field measurements. Following the HCM (2022) guidelines, the start-up lost time was taken as 2 seconds. The saturation headway was also set at 2 seconds, referencing the work of Duraku and Boshnjaku (2024). Additionally, yellow change intervals were set to 3 seconds to ensure intersection clearance safety.

For vehicle detection, inductive loop detectors with dimensions of 1.8m x 5m were used. These were placed approximately 2m from the stop line based on calculations for the distance ahead. The approach speed for the minor road (DOR approach) was calculated at 10 km/h, and vehicle spacing was set between 6 and 7m based on field observations.

Multiple simulations and observations showed that the minimum green time assigned was enough to clear all vehicles and satisfy passage time conditions. Since the traffic volume on the minor road was low, providing a unit extension only led to wasted green time. Therefore, no extension time was included in the final control logic.

**Input Parameters:**

The following parameters have been assumed based on field observations and HCM guidelines:

- Total Cycle Length (C) = 169 seconds
- Start-up Time Loss ($T_L$) = 2.0 seconds
- Saturation Headway (h) = 2.0 seconds/vehicle
- Distance from Detector to Stop Line (d) = 2.0 meters
- Vehicle Spacing (x) = 6 to 7 meters
- Approach Speed (S) of minor road= 10 km/h

**Minimum Green Time ($G_{min}$) Calculation**

Using Equation (4-2)

$$G_{min} = T_L + [h \times \text{Integer}(d/x)]$$
$$G_{min} = 2.0 + [2.0 \times \text{Integer}(2.0/6.5)]$$
$$G_{min} = 2.0 + [2.0 \times 0]$$
$$G_{min} = 2.0\text{seconds (minimum)}$$

However, based on FHWA recommendations for off-peak operations, the applied minimum green time is:

$$G_{min}\text{= 4 seconds}$$

**Passage Time (P) and Unit Extension (U) Calculation**

Using Equation (4-3):

$$P = d/(1.47 \times S)$$
$$P = 2.0/(1.47 \times 10) = 2.0/14.7$$
$$P = 0.136 \text{ seconds}$$

**Unit Extension (U) Selection**

No extension time was included in the final control logic.

**Maxmimum Green Time ($G_{max}$) Calculation**

Critical Traffic volume data for all phases is converted to Passenger Car Units (pcu/h) as follows, according to the HCM protocols:

| Phase | Volume in 15 mins | Factor | $V_{C_i}$ (pcu/h) |
|---|---|---|---|
| Phase 1 | 24.5 | 4 | 98 |
| Phase 2 | 250.75 | 4 | 1003 |
| Phase 3 | 46 | 4 | 184 |
| Phase 4 | 6.68 | 4 | 26.72 |

Total Critical Movement (VC):

$$\begin{aligned} V_C &= 98 + 1003 + 184 + 26.72 \\ &= 1311.72 \text{ pcu/h} \end{aligned}$$

Using Equation (4-4), the effective green time for Phase 4 is calculated:

$$\begin{aligned} g_i &= (C_i - L) \times (V_{C_i}/V_C) \\ g_i &= (169 - 4) \times (26.72/1311.72) \\ g_i &= 165 \times 0.02037 \\ g_i &= 3.36 \text{ seconds} \end{aligned}$$

**Cycle-by-Cycle Adjustment**

According to principles for enhancing controller flexibility to accommodate larger demands and enable cycle-by-cycle adjustments, the calculated green time is multiplied by an adjustment factor of 1.5:

$$g_i(\text{adjusted}) = 1.5 \times 3.36 = 5.04\text{seconds}$$

Hence, we adopted 5sec Maximum Green Time for minor roads.

### 4.9. Vehicle Actuated Programming

Vehicle Actuated Programming (VAP) is a capability within PTV Vissim that enables signal control strategies to be triggered based on the detection and actuation of vehicles, allowing the software to replicate programmable, actuated traffic signal operations. The control logic, stored in a .VAP file, is developed using the VisVAP (Visual VAP) tool, which allows users to design and represent the underlying signal control logic through a flowchart-based programming interface. Alongside this, an Interstage file (.pua) is also generated, which holds the VAP signal data. This interstage definition file specifies the signal groups, intergreen times, individual signal stages, and the interstage transitions. It is manually created and then imported into the VISSIM environment.

VisVAP enhances the use of freely definable signal control logics using the VAP language (Vehicle Actuated Programming) in offering a comfortable tool for creating and editing program logics as flow charts. The appearance and design of flow charts in VisVAP is similar to RiLSA 2010 (German de-facto law for signal controls) and has been enhanced to facilitate loops and other features. VisVAP can be used for both stage-based and signal group-oriented design.

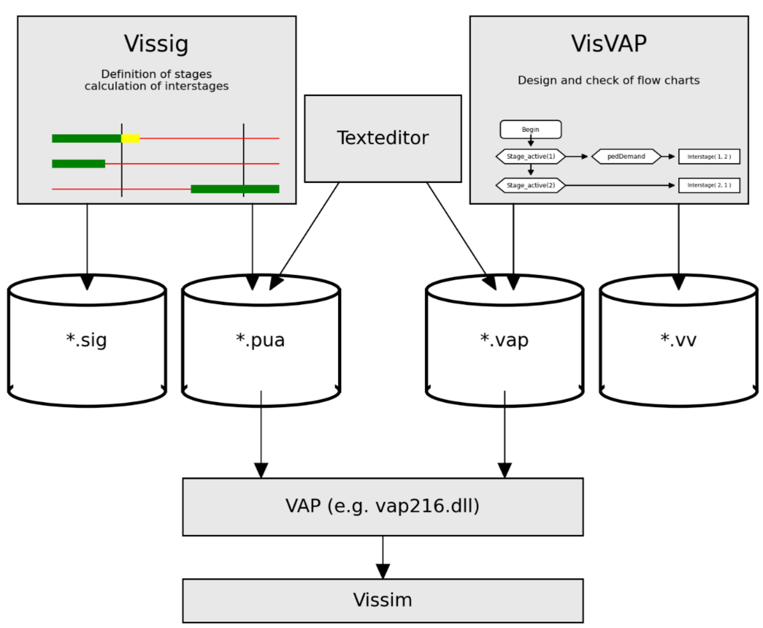


Figure 4-3. VisVAP Flowchart

Table 4-5. (PTV Planung Transport Verkehr AG, Karlsruhe, Germany)

| **Vap Functions** | **Meaning** |
|---|---|
| Interstage(<stage1>, <stage2>) | Runs an interstage |
| Interstage_active(<stage1>, stage2>) | Returns 1 if any interstage is active, otherwise 0. |
| Start_at(<timer>, <value>) | Starts the timer <timer> with starting value <value>. The first timer increment will take place in the next simulation second. |
| Interstage_duration(<stage1>, <stage2>) | Returns the current second of the interstage or 0 in case the interstage is not active. |
| Interstage_length(<stage1>, <stage2> ) | Returns length of interstage |
| Occup_rate(<no>) | Returns smoothened occupancy rate of detector <no> [0..1] |
| Occupancy(<no>) | Returns the elapsed time since detector <no> has been activated or 0 if no vehicle is present at end of time step. |
| P_interstage(<no>, <sec>) | Calls the interstage <no>, starting in its second <sec> (usually 0) |
| P_interstage_active(<no>) | Returns 1 if interstage <no> is active, otherwise 0 |

### 4.10. Model Developed According to SATSC Strategy

The Vehicle Actuated Programming (VAP) signal control has been applied to design the signal plan at this four-legged intersection. After setting up the necessary elements in the PTV Vissim interface, including lanes and connectors, vehicle volumes, traffic light configuration, and links, the block diagram logic was then implemented using the VisVAP tool. This intersection consists of four signal groups. Three of these signal groups operate as fixed signal groups, meaning their timing remains constant regardless of traffic conditions. These are:

SG1 - Right turn movement from the main lane

SG2 - Through movement of the main lane

SG3 - Art Gallery approach

SG4 - DoR approach

The fourth signal group, SG4, is assigned to the DoR approach and operates as a vehicle-actuated signal group. A detector is placed on this approach to sense the presence of incoming vehicles. When a vehicle is detected, it triggers the actuation of SG4, allowing the signal to respond dynamically to real-time traffic demand on that leg of the intersection.

This is the core application of VAP in this model, where the signal controller activates the DoR approach phase only when a vehicle is detected, thereby optimizing signal efficiency.

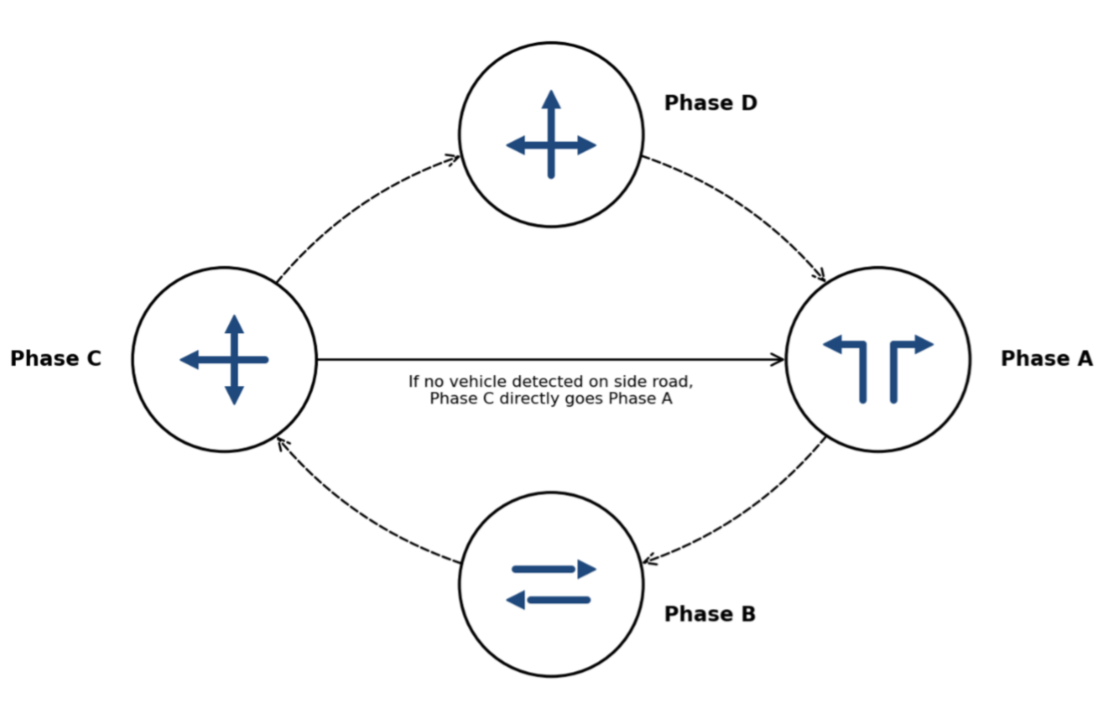


Figure 4-4. Block Diagram

The block diagram defining this control logic has been saved as a *.vap file, while the signal phases and interstage definitions have been exported as a *.pua file. Alongside the simulation model, analytical calculations have also been performed to validate the signal design at this intersection.

The logic files (*.pua and *.vap) are then imported to the Vissim model in the signal control menu. The signal controller window in Vissim is as shown in Figure 4-5

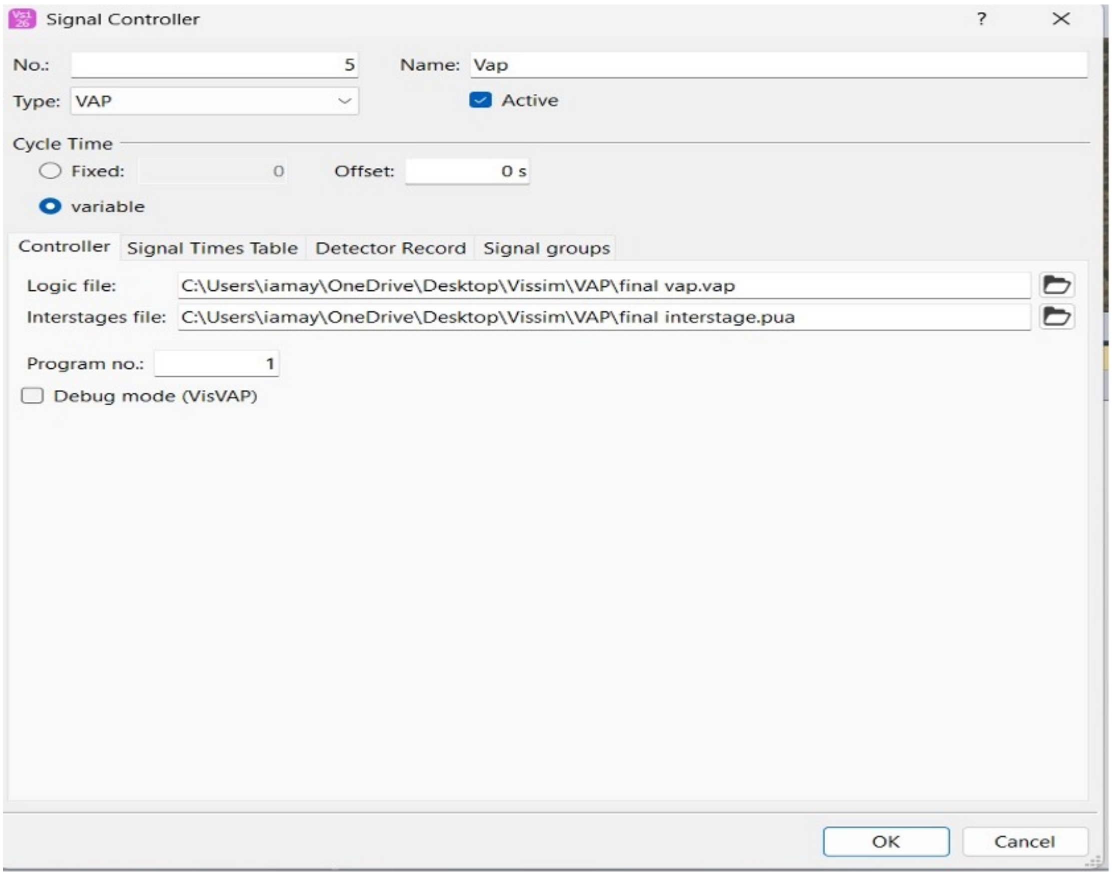


Figure 4-5. Signal controller Window

The inductive loop detector, with dimensions of **1.8** × **5 m,** has been positioned at an optimal distance of approximately d = 2 m from the stop line on the DoR approach, determined through distance-ahead calculations. The proposed vehicle speed on the DoR approach is set at **S = 40 km/h**, and the unit extension duration is set as **U = 3 seconds**, which is greater than the passage time **P = 0.57 seconds** from the detector to the stop line, in accordance with standard recommendations. The cycle duration C at the Babar Mahal Intersection varies dynamically depending on the traffic demand on the DoR approach. When no vehicles are present on the DoR approach, only the three fixed signal groups (SG1, SG2, and SG3) are activated, and the minimum cycle is set to Cmin = 64seconds. The time that would have been allocated to the DoR approach is directly transferred to the main road fixed phases, maximizing their operational efficiency.

When vehicles are detected on the DoR approach, the full signal cycle is activated, engaging all four signal groups, including the actuated SG4, with the cycle duration Cmax = 72 seconds, accommodating traffic demand from all approaches of the intersection. The signal plans of the SATSC strategy are shown in Figure 4-6

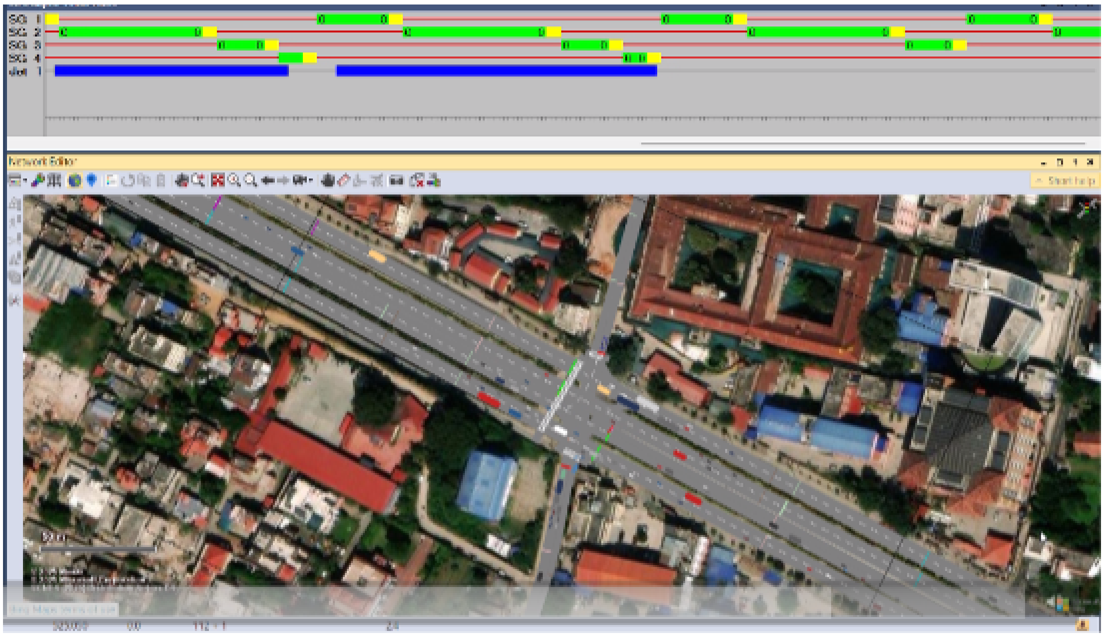

Figure 4-6. Signal plans for SATSC

## 4.11. Logic Design Controls

The VAP control logic for the Babar Mahal Intersection was implemented in VisVAP as a flowchart governing four signal groups, where SG1, SG2, and SG3 operate as fixed signal groups, and SG4 operates as a vehicle-actuated group on the DoR approach.

Initialization: The program starts by evaluating Init = 0. When true, TimerMain is started, and the flag is set to Init:= 1, ensuring initialization occurs only once.

Stage 1 to Stage 2: When Stage_active(1) AND TimerMain ≥ 15, the controller triggers Interstage(1,2) and resets TimerMain to zero, transitioning from the main lane through movement to the right turn phase.

Stage 2 to Stage 3: When Stage_active(2) AND TimerMain ≥ (15 + Interstage_length(1,2)), the controller triggers Interstage(2,3), transitioning to the Art Gallery approach phase.

Stage 3 to Stage 4 (Actuated): When Detection(1) AND TimerMain ≥ Interstage_length(2,3) AND Stage_active(3), indicating vehicle presence on the DoR approach, Interstage(3,4) is triggered, activating SG4. TimerSub starts, and TimerMain stops.

Stage 4 to Stage 1: When TimerSub ≥ Interstage_length(3,4) AND Stage_active(4), the controller triggers Interstage(4,1), returning to Stage 1, restarting TimerMain and stopping TimerSub.

**No Detection case:** When Stage_active(3) AND TimerMain ≥ Interstage_length(2,3) with no vehicle detected, Stage 4 is bypassed, and Interstage(3,1) is triggered directly, maintaining the minimum cycle of Cmin = 64 seconds with only the three fixed signal groups active.

# 5. INPUT ANALYSIS

## 5.1. Study Area

## 5.2. Data Collection

## 5.3. Volume Data in flow Diagram

## 5.4. Average Flow Data in Graphical Form

## 5.5. Input in PTV Vissim

## 5.6. Input in VISVAP

# 6. COMPUTATION

## 6.1. Traffic Flow Calculation

## 6.2. Calculation of Saturation Flow

## 6.3. Calculation of Percentage of Turning Movement

## 6.4. Calculatin of Percentage of Heavy Vehicles

## 6.5. Computation of Peak Hour Factor(PHF) and Adjusted Volume

## 6.6. Calculatin of LOS Using V/C Ratio

## 6.7. Improvement Analysis

### 6.7.1. Signal Optimization

### 6.7.2. Comparision of Intersection Parameters

# 7. RESULTS AND DISCUSSION

# 8. CONCLUSION AND RECOMMENDATION

## 8.1. Conclusion

## 8.2. Recommendation

## A. ANNEX I

## REFERENCES